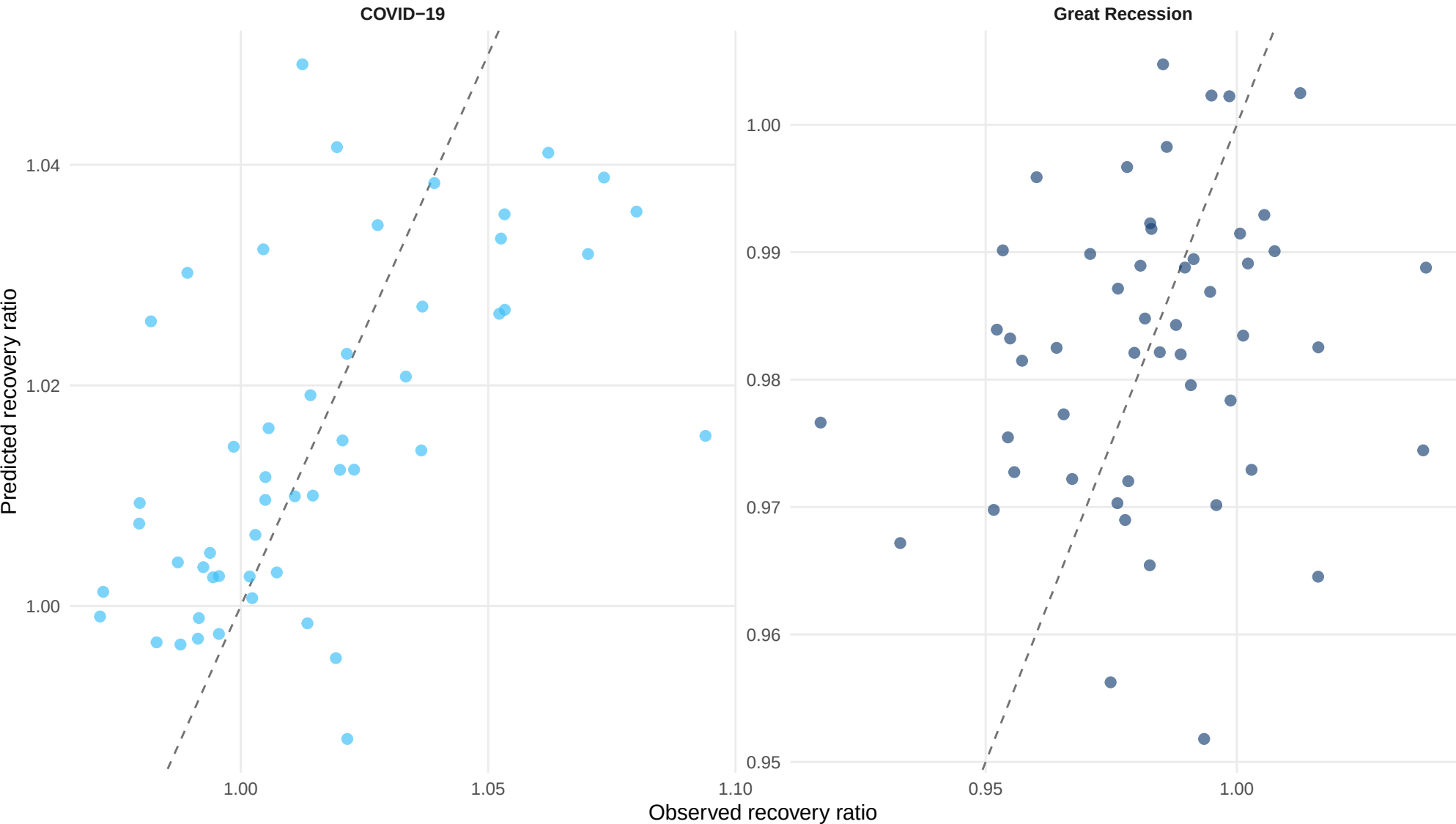


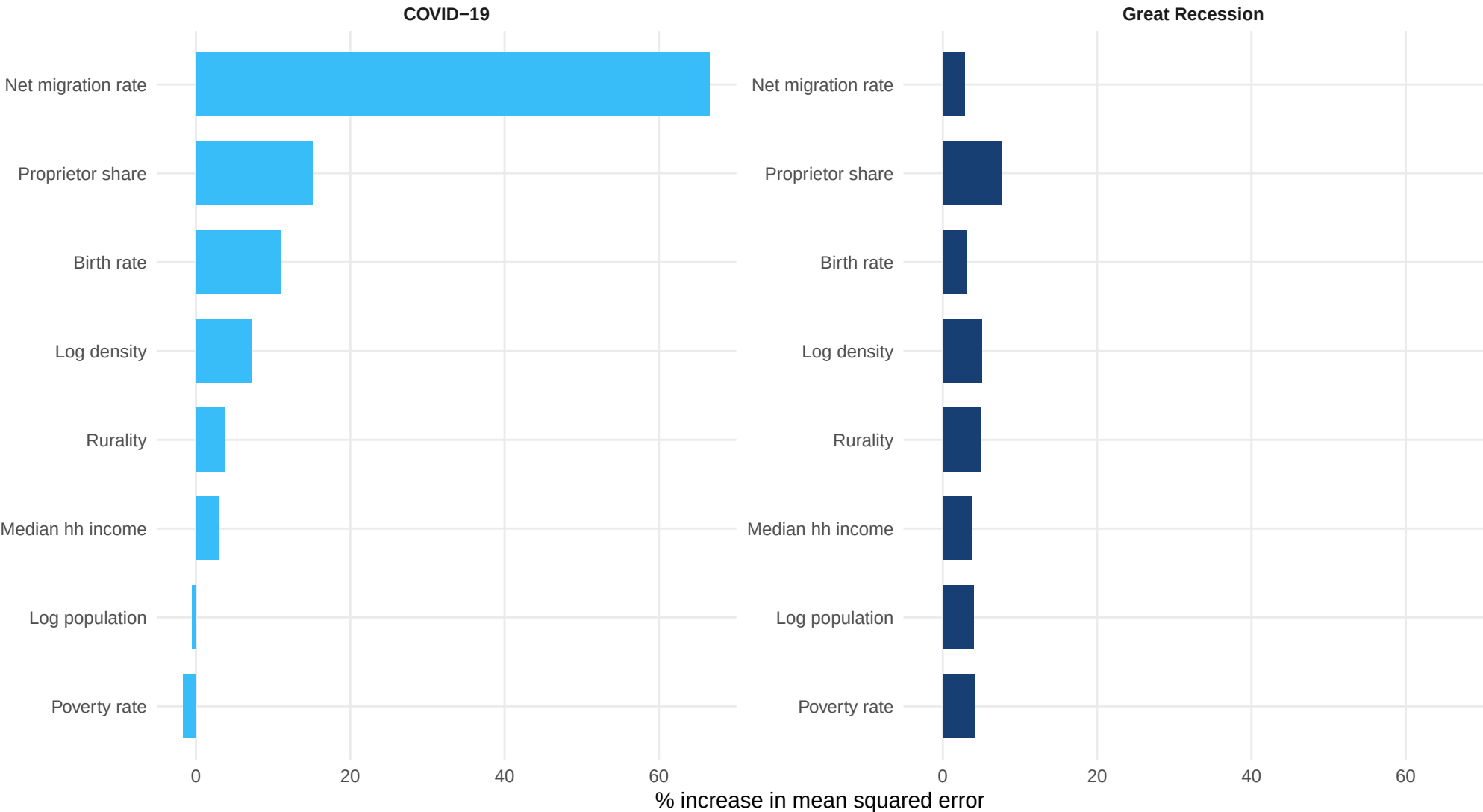
State random forest: observed versus predicted

Predictions are out-of-bag estimates for the 51 state/DC observations



State random-forest variable importance

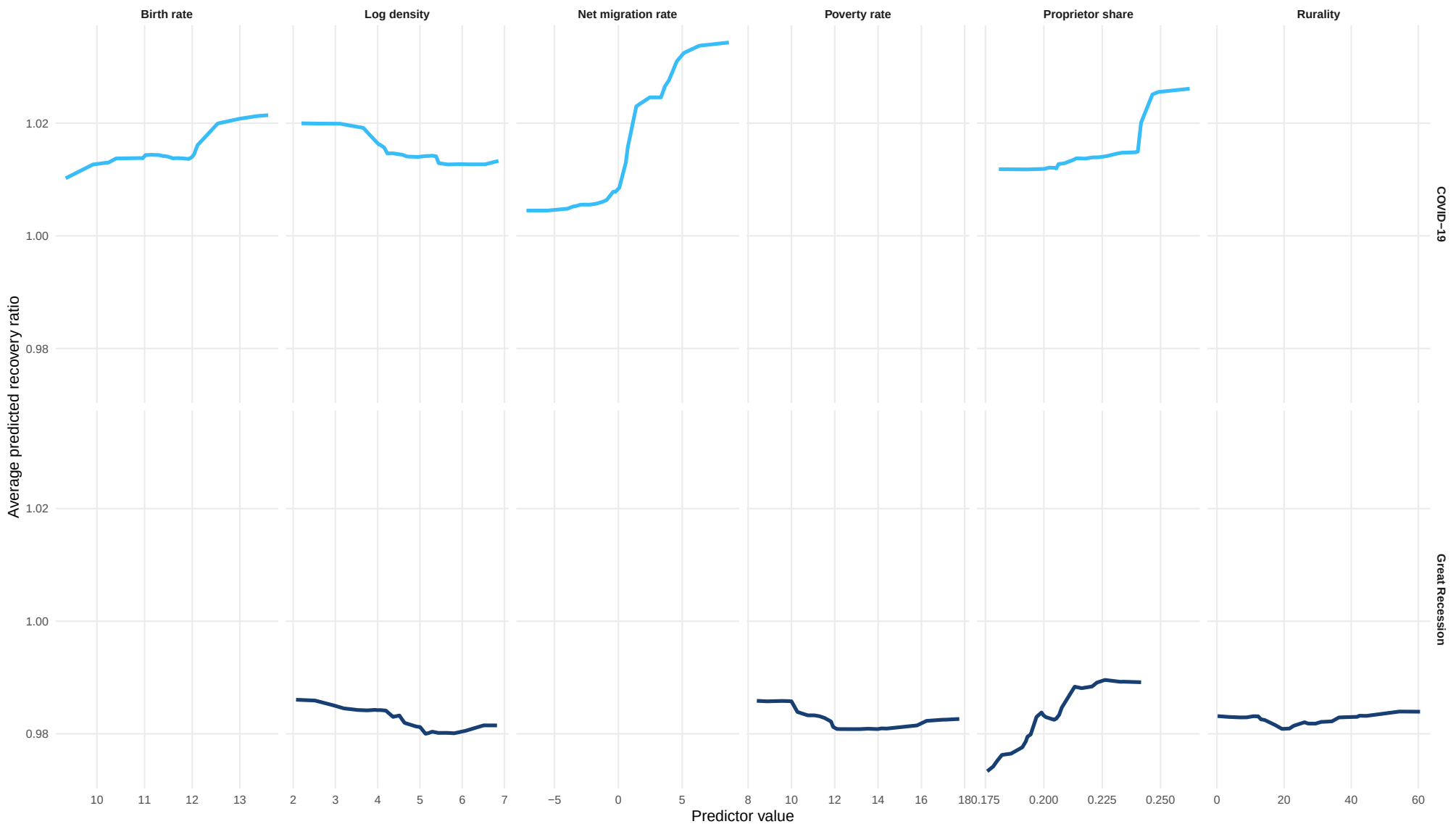
Permutation importance: accuracy loss when each predictor is shuffled



Importance indicates predictive reliance, not direction or causality.

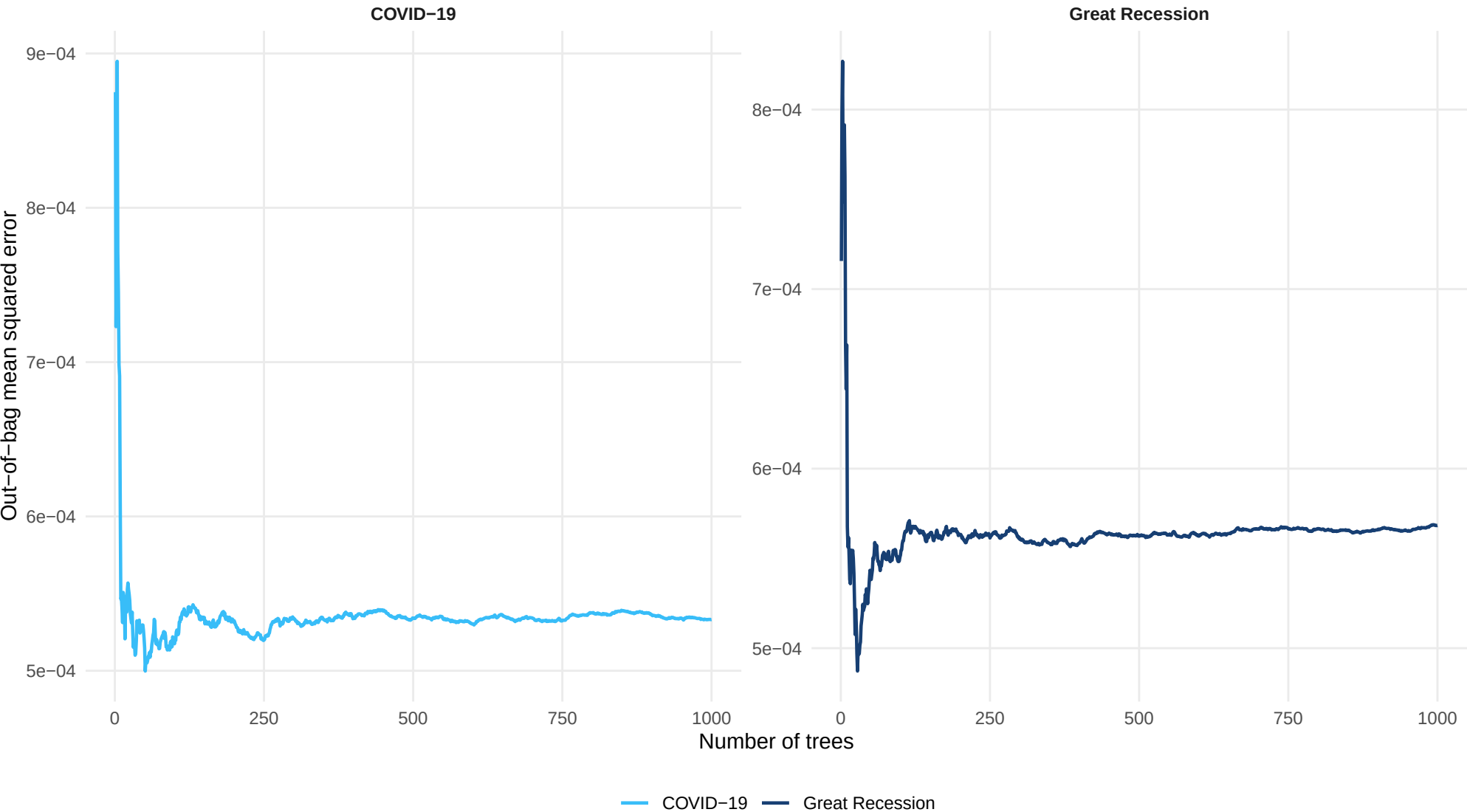
State random-forest response profiles

Marginal prediction curves for the most important numeric predictors



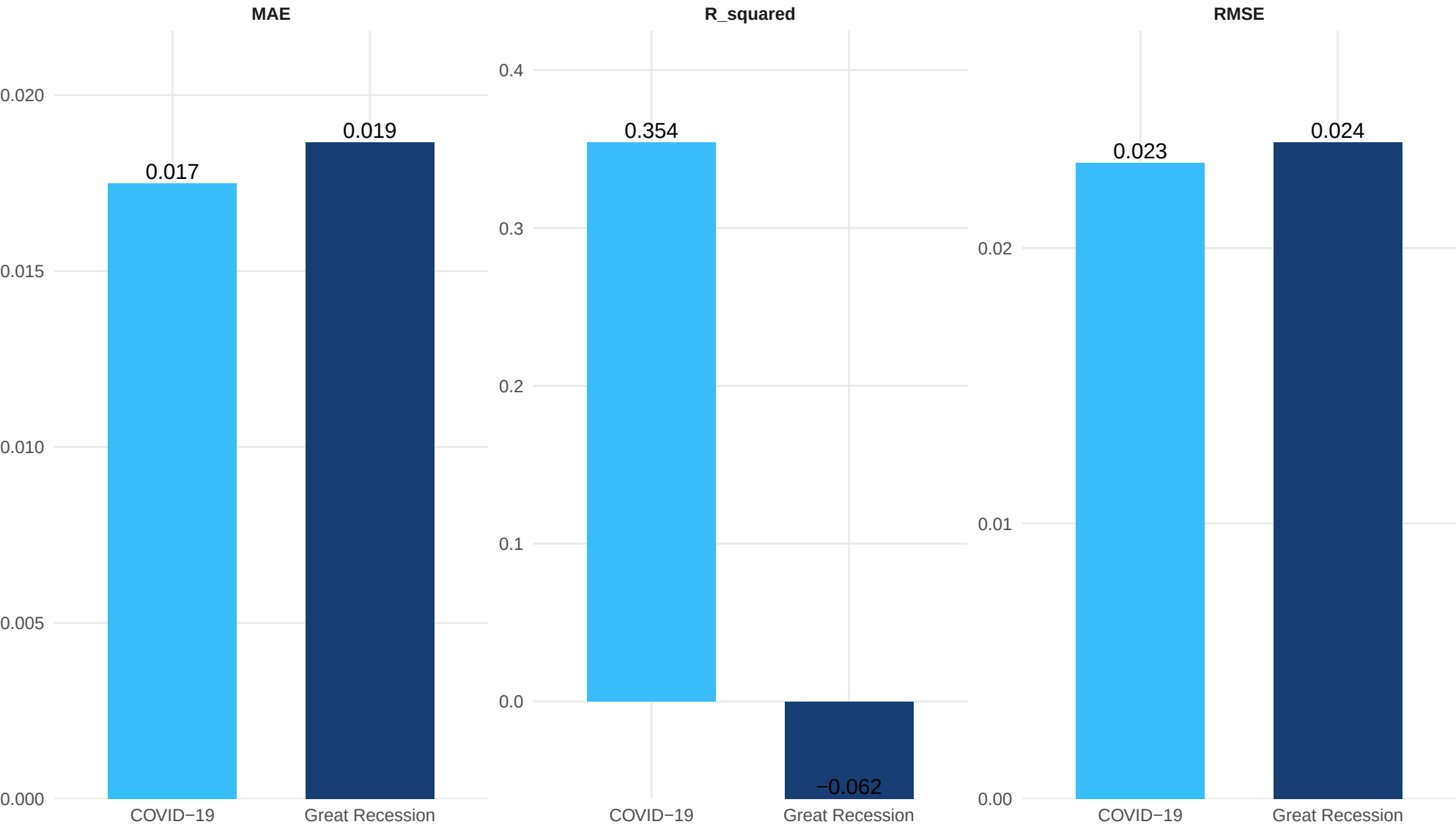
State random-forest convergence

Out-of-bag error should stabilize as trees are added



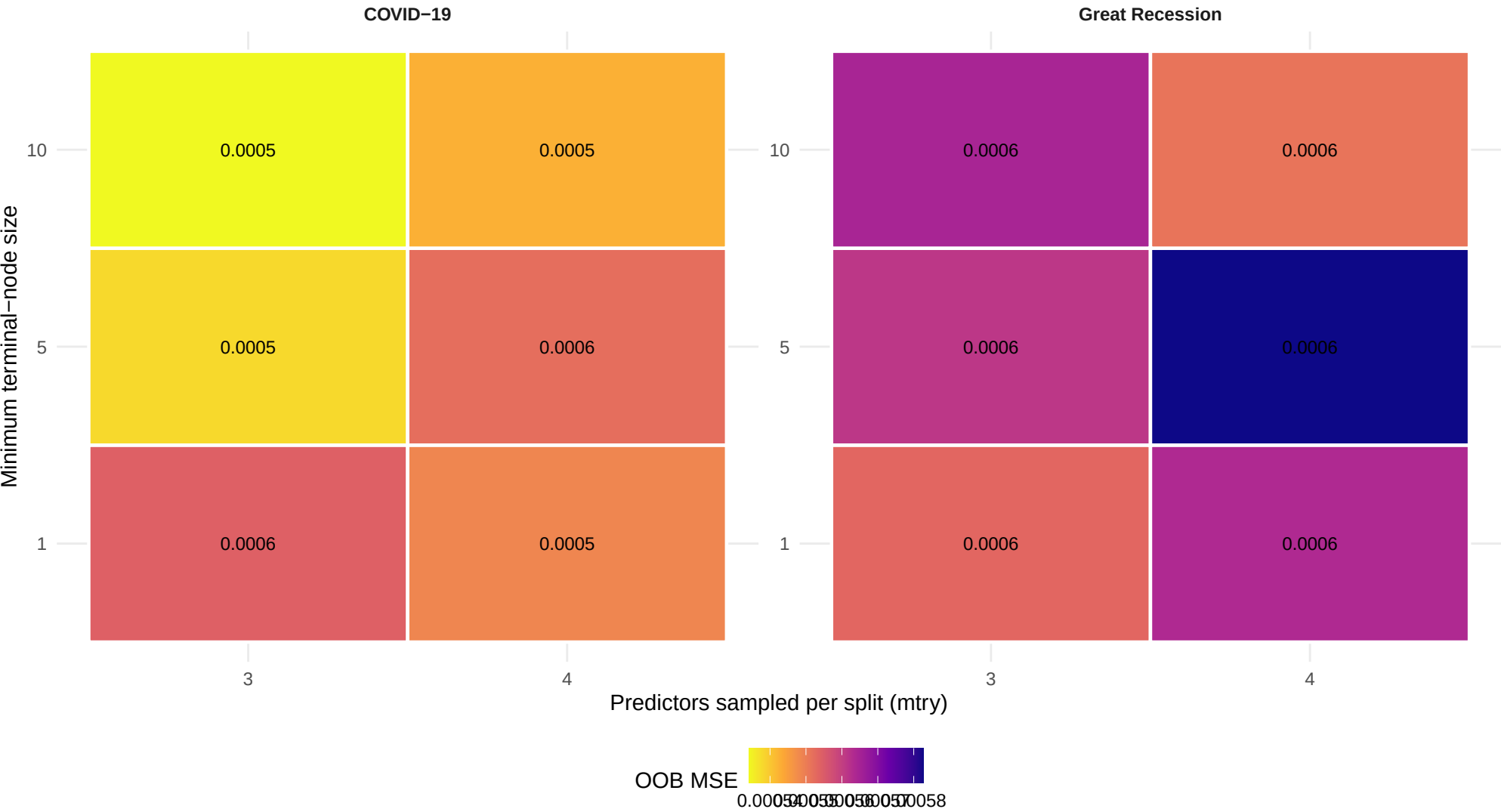
State random-forest performance

Lower RMSE and MAE are better; higher R-squared is better



State random-forest hyperparameter search

Training-set out-of-bag MSE for combinations of mtry and node size



The final model uses the lowest-OOB-error combination.